

JavaScript Practice Tasks

Basic • Intermediate • Advanced

Prepared for Students by CodeJavid

These tasks are designed to strengthen real JavaScript fundamentals: constructor functions, this keyword, prototypes, objects, arrays, functions, prototype inheritance, custom polyfills, and deeper JavaScript behavior.

Practice by writing the code yourself without copy-pasting solutions.

BASIC LEVEL TASKS

#	Task
1	Create a Car constructor function with brand and model properties.
2	Add a getCarName() method using this keyword.
3	Compare primitive string vs new String().
4	Compare primitive number vs new Number().
5	Create a function using new Function().
6	Create arrays using [] and new Array().
7	Create objects using {} and new Object().
8	Print this inside an object method.
9	Print global this and observe behavior.
10	Create a Student constructor and create 3 student objects.

INTERMEDIATE LEVEL TASKS

#	Task
1	Add getResult() using Student.prototype.

2	Understand memory optimization using prototype methods.
3	Create <code>Array.prototype.myLastElement()</code> .
4	Create <code>String.prototype.reverseText()</code> .
5	Inspect constructor property of objects.
6	Explore prototype chain using <code>__proto__</code> and <code>Object.getPrototypeOf()</code> .
7	Build a Bank Account system using prototypes.
8	Compare arrow function vs normal function this behavior.
9	Build a Calculator constructor with operations.
10	Create a simple login system using constructor functions.

ADVANCED LEVEL TASKS

#	Task
1	Implement the new keyword manually.
2	Create custom <code>Array.prototype.myMap()</code> .
3	Create custom <code>Array.prototype.myFilter()</code> .
4	Create custom <code>Array.prototype.myReduce()</code> .
5	Implement prototype inheritance with Animal and Dog.
6	Build a mini Event Emitter system.
7	Create a deep clone function.
8	Implement <code>Function.prototype.myBind()</code> .
9	Create a class system using ES5 constructors and prototypes.
10	Build a tiny React-like state system.

Important:

If you master:

- this keyword

- prototype chain
- functions
- closures
- async behavior
- memory/reference

Then learning React, Next.js, Node.js, or AI frameworks becomes much easier.